



BLADE RUNNER

South African athlete Oscar Pistorius lost the lower part of both his legs when he was a baby. He now walks and runs on J-shaped artificial feet made of carbon fiber. These are so good that Pistorius, nicknamed “Blade Runner” by his fans, is able to compete against able-bodied athletes, as in the 2011 World Athletics Championships, shown below.



▲ **EAR IMPLANT** A cochlear implant picks up sound and transmits it to the inner ear, giving some sense of hearing to people who are deaf.

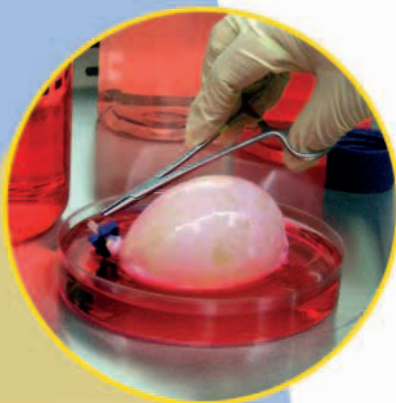


A computer controls the motors that make the arm and hand move.

◀ **ROBOTIC ARM** This high-tech mechanical arm is controlled by thought alone. Sensors on the shoulder pick up nerve signals inside the wearer's body and relay them to a computer, which controls the arm's movements.

A motor inside the elbow makes the arm bend.

PACEMAKER ▲ Implanted into the chest wall, a pacemaker sends an electric signal along a wire to the heart, controlling the rate at which it beats.



◀ **NEW BLADDER** The bladder is one of the few organs that scientists can grow in a lab, using a patient's own body cells.

ORGAN TRANSPLANTS

The most common way of replacing a damaged internal organ is to use a healthy organ from a donor—an organ transplant. Transplants can save lives and make people healthy again, but recipients must take powerful drugs to stop their bodies from rejecting the new organs.



▲ **KIDNEY TRANSPLANT** Surgeons prepare a donated kidney before implanting it into a recipient.